S_3, P_1:

a) $\binom{49}{6} = 13\,983\,816$

b) I: $6/6 = 1$

II $5/6 = \binom{6}{5} \cdot \binom{43}{1} = 258$

III $4/6 = \binom{6}{4} \cdot \binom{43}{2} = 13\,545$

IV: $3/6 = \binom{6}{3} \cdot \binom{43}{3} = 246\,820$

S_2, P_6:

a) 1. Each line has one rook

Put the rooks one by one (top to bottom & line by line)

$$8 \cdot 7 \cdot 6 \cdot \ldots \cdot 1 = 8!$$

b) Place the rooks one by one

$$\underset{8^2}{W} \rightarrow \underset{7^2}{B} \rightarrow \underset{6^2}{B} = (8 \cdot 7 \cdot 6)^2$$

c) $\dfrac{(8 \cdot 7 \cdot 6)^2}{3!}$ d) $\dfrac{(n^{\underline{k}})^2}{k!}$

↳ 1. Choose the lines

k lines of $n = \binom{n}{k}$ options

2. Place the rook 1 by 1 from top to bottom

$n \cdot (n-1) \cdot \ldots \cdot (n-k+1) = n^{\underline{k}}$

$\Rightarrow \binom{n}{k} \cdot n^{\underline{k}}$

6) $a+b+c+d = 10 \quad a \geq 1$

$a \neq 9$

$$\begin{array}{r} 26 \cdot \\ 11 \\ \hline 26 \\ 26 \\ \hline 286 \end{array}$$

b) $a+b+c+d =$

9) ① ② ③ ... ⑪ ⑫

__ __ __ __

$1 \leq x_1 < x_2 < x_3 < x_4 \leq 12 \quad a < b \ (b-a \geq 2)$

$1 \leq x_1 < x_2 - 1 < x_3 - 2 < x_4 - 3 \leq 9$

$y_1 \quad y_2 \quad y_3 \quad y_4$

$x = (2, 5, 7, 12) \qquad y = (4, 5, 8, 9)$

$y = (2, 4, 5, 9) \qquad x = (4, 6, 10, 12)$

y can be obtained $= \binom{9}{4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4} = 126$